



Rupanjan Bhattacharyya

Roll No.: 2023EEB106

Bachelor of Technology [Four-Year Degree Course]

Electrical Engineering

Indian Institute of Engineering Science and Technology, Shibpur

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GitHub Profile

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EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur

Bachelor of Technology, Electrical Engineering

6th Semester 2026

CGPA: 8.07/10 | GATE EE All India Rank: 4613

DAV Model School, IIT Kharagpur

Central Board of Secondary Education

AISSE 2023

Percentage: 92.6%

DAV Model School, IIT Kharagpur

Central Board of Secondary Education

AISSE 2021

Percentage: 97%

RESEARCH EXPERIENCE

NextGen Compute and Storage Lab, IIT Madras

Summer Fellow

May 2026 – Present

Chennai, India

- Derived a physics-based compact model for the voltage-dependent capacitance of hafnium-oxide resistive memory devices by reformulating filament-gap dynamics as a chemical-kinetics rate balance.
- Implemented the model in Verilog-A, enabling CMOS image sensor pixel simulations that improve dynamic range by 40% over a baseline pixel and 25% over prior hybrid designs.

Nano Intelligence Lab, Centre for Nanotechnology, IIT Roorkee

Research Intern

Mar 2026 – Present

Roorkee, India

- Mapped a six-head Vision Transformer for CIFAR-10 image classification onto a cascaded memristive crossbar attention datapath, executing 98.7% of multiply-accumulate operations in memory and developed a memristor transformer accelerator as a physically-consistent digital twin.
- From a device-level non-ideality analysis, then designed and implemented a mitigation pipeline that affine per-crossbar IR-drop calibration, replica-based cell redundancy, time-aware retention-drift scaling, and target-sparsify stuck-at-fault mapping recovering accuracy from 34% to 72% under 3Ω line resistance and from 35% to 87.5% after a 10^4 s retention interval at 85°C , and benchmarked the resulting digital twin against a fabricated perovskite transformer engine to propose seven transferable hardware design principles.

Centre of Excellence for Intelligent Sensors and Systems, CSIR-CSIO

Summer Research Intern

May 2025 – July 2025

Chandigarh, India

- Engineered a four-channel, 24-bit real-time seismic acquisition and classification system on a Raspberry Pi, sustaining 240 hertz per-channel sampling with timing jitter under five microseconds.
- Built a 47-feature multi-domain detection pipeline with cross-channel event correlation and adaptive batch processing, achieving over 87% real-time classification accuracy with under 3-millisecond inference latency.
- Reduced false-positive event detections by approximately 40% by designing a 3-out-of-5 sample voting mechanism layered on a recursive energy-ratio trigger algorithm.
- Designed a four-channel seismic-acquisition circuit board in KiCad with instrumentation amplifiers and Butterworth low-pass filters, isolating analog and digital grounds through a dedicated power converter with diode-based over-voltage protection.

Indian Institute of Technology, Guwahati

Winter Research Intern

Dec 2024 – Jan 2025

Guwahati, India

- Architected a computational imaging system for classroom attention-level detection, designing two parallel convolutional neural networks to jointly process facial emotion and head pose from a single video stream.
- Developed a weighted fusion algorithm combining emotion and head-pose signals into a real-time attention score, and explored model quantization for edge deployment.

ACADEMIC PROJECTS

•Transmission Line Insulator Fault Detection using PINN

Dec 2025 – Feb 2026

Skills: Physics-based Modeling, Numerical Simulation, Power System Design, Deep Learning

- Architected a cascaded convolutional neural network and Physics Informed Neural Network (PINN) pipeline that isolates insulators at 87.8% intersection-over-union before localizing thermal faults.
- Achieved 95.4% mean average precision for defect detection by fusing convolutional segmentation with a physics-informed heat-source solver, cutting physically implausible false positives.
- Generated fault severity scores and maintenance-priority flags from Monte Carlo dropout uncertainty estimates on the PINN's predicted heat-source intensity.

•Electromagnetic Levitation - Physics-Based Digital Twin

Aug 2024 – Nov 2024

Skills: Ansys Electronics, FEMM, Python, Control Systems

- Simulated a repulsive eddy-current levitation actuator in Ansys Maxwell and FEMM using three-dimensional transient solvers and adaptive meshing to compute magnetic field distribution and net lift force.
- Modeled the system as a nonlinear state-space digital twin in Python, predicting a 25.6 radian-per-second natural frequency and optimizing coil geometry to match the target force constant within 1%.

ACADEMIC AND PROFESSIONAL DEVELOPMENT

•NPTEL Digital Design with Verilog

May 2026

- Procured Elite certification in this 12-week course conducted by Indian Institute of Technology Guwahati.

•Samsung Grade-I Fellowship under India Semiconductor Workforce Development Program

October 2025

- Gained hands-on experience in practical semiconductor device modeling using Synopsys Sentaurus TCAD's Structural Device Editor (SDE) scripting, Sprocess to simulate fabrication flows, and model calibration, ensuring simulation results align with experimental data.

•Texas Instruments' Analog System Lab Kit Pro Hardware Design Contestant

October 2025

- Participated in the Texas Instruments' Hardware Hackathon using the ASLK Pro (Analog System Lab Kit) to design and test analog signal processing circuits.

•Texas Instruments' Build Your Technical Edge – Analog Certification

July 2025

- Completed this workshop, gaining hands-on exposure to Analog System Design, Simulation flows, and industry-standard testing.

HONORS AND AWARDS

•Winner of National Engineering Innovation Challenge organized by WBPDCCL

February 2026

- The National Engineering Innovation Challenge organized by the West Bengal Power Development Corporation involved a national level competition among young engineers across India to design impactful AI-driven innovations addressing real-world challenges in Healthcare and Power Systems. The prototypes were evaluated by a panel of eminent academic and industry experts.

SCHOLARSHIPS AND FELLOWSHIPS

•Exide Diksha Scholarship

February 2026

- A merit-based scholarship program curated as part of its Corporate Social Responsibility (CSR) initiatives, following a structured selection process that included project evaluation and personal interviews, 24 students were chosen as the first cohort of Diksha Scholars.

TECHNICAL SKILLS AND INTERESTS

- **Signal Processing & Machine Learning Frameworks:** Python (NumPy, Pandas, Matplotlib, TensorFlow), C/C++ (Embedded Backend), Data Preprocessing, Feature Extraction Pipelines
- **Simulation & Digital Twin:** Keysight ADS, Cadence Virtuoso, AMD Vivado, Sentaurus Workbench (SDE scripting and Sprocess), MATLAB, Ansys Electronics, SolidWorks, AutoCAD, KiCAD, LTspice, QUCS
- **Edge Computing & Real-Time Systems:** Hardware-Software Co-Design, Multi-sensor Fusion, Edge Inference, Real-Time Processing, C for Microcontrollers, Verilog, Arduino IDE, Parallel Processing Architectures
- **Relevant Coursework:** Signals and Systems, Analog & Digital Electronics, Microprocessors & Micro-controllers, Transducers and Instrumentation, Network Theory, Control Systems, Electrical Machines, Electric Drives, Classical Electrodynamics, Electrical and Electronic Measurements, Power Electronics, Power Systems, Mathematics, Physics, Mechanics, Introduction to Computing, Chemistry
- **Areas of Interest:** Computer Architecture, Neuromorphic Computing, Signal Processing, Deep Learning, Embedded Systems, VLSI Design, Control Systems, Robotics and Automation